Books.

23-07-2024
Para Inf

1. Statistical Inference — Casella-Berger
2. Mathematical Statistics — Bickel-Doksum
3. Introduction to prob & stat — Rohatgi-Saleh
4. Theory of Point Estimation - Eric Lehmann
5. Testing Statistical Hypothesis - Lehmann-Romano

Background

Observe: data x is realization of r.v. X (height/weight) has some distribution with CDF F

If we repeat this experiment (of selecting students randomly & checking their heights) we obs. $x_1, \dots, x_n$ which is a realization of r.v. $(X_1, \dots, X_n)$ with x_i being the realization of $X_i \sim F$

We say $(X_1, \dots, X_n)$ is a "sample" from distribution/population F. In addition, if obs. are taken independently then we say $X_1, \dots, X_n$ is a random sample from F. $X_1, \dots, X_n \sim F$

Statistic: Any function $T(X_1, \dots, X_n)$ is called a statistic if T does not involve any unknown parameters

ex. $\overline{X}$, sample median $\tilde{X}$, S^2, ...

Sampling Distribution Suppose $X_1, \dots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

(1) $\dfrac{\sqrt{n}\,\overline{X}-\mu}{\sigma} \sim N(0,1)$

$S = \sqrt{\frac{1}{n-1}\sum_i (X_i - \overline{X})^2}$

(2) $\dfrac{\sqrt{n}(\bar{X}-\mu)}{S} \sim t_{n-1}$

(3) $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$, $\bar{X} \perp\!\!\!\perp S^2$

$$X^{p\times 1} \sim N_p\left(\mu^{p\times 1}, \Sigma^{p\times p}\right)$$

$$l'X \sim N(l'\mu, l'\Sigma l) \;\; \forall\, l \in \mathbb{R}^p$$

$$Z \sim N_p(0, I_p)$$

$$\Rightarrow \mu + \Sigma^{1/2} Z \sim N_p(\mu, \Sigma)$$

Statistical Inference

Let $X_1, \dots, X_n$ be a sample from some distribution F

Goal: to learn about F.

Inference

- Parametric (Semiparametric)
  - Assume, we know the form of F_θ where $\theta \in \mathbb{R}^k, k \in \mathbb{N}$. Only unknown is $\underline{\theta}$, non-random
    - Estimation
      - Point
      - Interval/Set
    - Hypothesis Testing
- Non-parametric
  - The form of F is not known
    - Estimation
      - Point
      - Interval/set
    - Testing
- Bayesian
  - Unknown θ is random variable with some dist^{bn} $\underbrace{\pi(\theta)}_{\text{prior}}$

Parametric Inference

Setup: Suppose we have sample $X_1, \ldots, X_n$ from some dist^{bn} $\{F_\theta : \theta \in \Theta\}$ $\downarrow$ model
where Θ = parameter space = all possible values of θ

goal: Learn about θ

Ex. $Ber(p)$: $p \in (0,1)$

$N(\mu, \sigma^2)$ $\quad \theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}$

$X \sim N(|\theta|, 1)$??

"Not Identifiable"

A probability model is called "identifiable" if $\theta_1 \neq \theta_2$
$\Rightarrow F_{\theta_1} \neq F_{\theta_2}$ $\forall \theta_1, \theta_2 \in \Theta$

$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $\varepsilon_{ij} \sim N(0,1)$
$j = 1, 2, \ldots, n_i$ $\quad i = 1, 2, \ldots$

$\longrightarrow \underset{\sim}{\theta} = (\mu, \alpha_1, \alpha_2)$

$\underset{\sim}{\theta}' = (\mu + c, \alpha_1 - c, \alpha_2 - c)$

So $\alpha_1 + \alpha_2 = 0$ Assumed

$\underset{\sim}{\theta} \neq \underset{\sim}{\theta}' \Rightarrow F_\theta = F_{\theta'}$

Point Estimation: $\underset{\sim}{X} = (X_1, \ldots, X_n)$ has pmf/pdf $f_\theta(x) : \theta \in \Theta$
$\mathcal{X}$ = sample space = set of all possible values of $(X_1, \ldots, X_n)$

A statistic $T : \mathcal{X} \longrightarrow \Theta$ is called a point estimate of θ

Ex. $N(\mu, \sigma^2)$: $\theta = (\mu, \sigma^2)$
A point estimator of (μ, σ^2) is $(\bar{X}, S^2)$

$T_1 = (\bar{X}, S^2) \quad T_2 = (\tilde{X}, \frac{1}{n}\sum |X_i - \tilde{X}|)$

Estimation Methods.

1. Method of Moments. (MME):

Equate sample moments with population moment (estimates)

Disadvantages

(i) Estimate may not be in parameter space

(ii) May be biased

(iii) Moments may not exists

Advantages

(i) Moments, if exists, match.

2. Maximum Likelihood Estimator (MLE):

$$X_1, \dots, X_n \sim f(x|\theta) \quad \theta \in \Theta$$

$$\mathcal{L}(\theta|\underset{\sim}{X}) = f(\underset{\sim}{X}|\theta)$$

$$\hat{\theta}_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}} \, \mathcal{L}(\theta|\underset{\sim}{X})$$

Advantages

(i) $\hat{\theta}_{MLE} \in \Theta$

(ii) Invariance Principle: Let $X = (X_1, \dots, X_n) \sim f(x|\theta)$, $\theta \in \Theta$ and $h: \Theta \to \mathbb{R}^k$ for some k

If $\hat{\theta}_{MLE}$ is the MLE of θ, then $h(\hat{\theta}_{MLE})$ is MLE of $h(\theta)$.

Under regularity conditions

1) $\hat{\theta}_{MLE} \xrightarrow{P} \theta$

2) $\hat{\theta}_{MLE} \xrightarrow{a.s.} \theta$ (strong assumption)

3) $\sqrt{n}(\hat{\theta}_{MLE} - \theta) \sim \mathcal{N}(0, I(\theta)^{-1})$

$X_1, \dots, X_n \overset{iid}{\sim} \exp(\mu, \sigma)$

$$f(x|\mu,\sigma) = \frac{1}{\sigma} e^{-\frac{(x-\mu)}{\sigma}} \mathbb{1}(x \geq \mu), \quad \sigma > 0$$

$$\Theta = \mathbb{R} \times \mathbb{R}^+$$

But $\max\limits_{\theta_1,\theta_2} \mathcal{L}(\theta_1, \theta_2 | \underset{\sim}{x}) \neq \max\limits_{\theta_1} \max\limits_{\theta_2} \mathcal{L}(\theta_1, \theta_2 | \underset{\sim}{x})$ in general

But Fix θ_1 solve $\max\limits_{\theta_2} \mathcal{L}(\theta_1, \theta_2 | \underset{\sim}{x}) \to \hat{\theta}_2$ maximizer

If $\hat{\theta}_2$ does not depend on θ_1, then it works.

Choose $\theta_1 = \sigma$, $\theta_2 = \mu$

$\hat{\mu} = X_{(1)}$, $\sigma \to$ ✓